What you plotted are the iterated discrete integrals ("prefix sums") of the sign of the Thue–Morse sequence

$$\tau(n) = (-1)^{s_2(n)} \qquad (s_2(n) = \text{sum of binary digits}).$$

Let $S^{(0)}(n) = \tau(n)$ and for $k \geq 1$

$$S^{(k)}(n) = \sum_{j=0}^{n} S^{(k-1)}(j) \qquad (n \geq 0)$$

(the k-fold accumulation). The lists you printed are $S^{(0)}, S^{(1)}, \ldots$.

Below is a clean way to see *why these shapes quickly become the Fabius function* (after a simple rescaling), and where the runs of zeros come from.

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1) Self-similarity in the data, and the Fabius equation

The Thue–Morse signs obey

$$\tau(2n) = \tau(n), \qquad \tau(2n+1) = -\tau(n), \qquad \tau(2n) + \tau(2n+1) = 0.$$

That last identity explains the second row of your output: every odd partial sum is zero.

Define, for each k, the *normalized, piecewise linear* function on $[0, 1]$

$$F_k(x) := \ 2^{-\binom{k}{2}}\, S^{(k)}(\lfloor 2^k x \rfloor) \quad \text{and join consecutive grid points } x_j = j/2^k. \tag{1}$$

A one-line computation using $\Delta S^{(k)}(n) = S^{(k-1)}(n)$ gives the *exact discrete functional equation*

$$2^k (F_k(x_{j+1}) - F_k(x_j)) = 2\, F_{k-1}(2x_j + \tfrac{1}{2^{k-1}})\,. \tag{2}$$

As $k \to \infty$, the mesh $h = 2^{-k} \to 0$ and the right–hand side becomes $2F(2x)$, where F is the (unique) C^∞ function on $[0, 1]$ with

$$F'(x) = 2F(2x), \qquad F(0) = 0,\ F(1) = 1, \tag{3}$$

i.e. the **Fabius function**. Equation (2) is therefore a consistent first-order finite-difference scheme for (3). That is the conceptual reason your rescaled curves increasingly look like F.

How to rescale. The power $2^{\binom{k}{2}} = 2^{k(k-1)/2}$ you noticed numerically (e.g. the 2^{15} plateau for $k = 6$) is exactly the factor needed to keep the vertical scale bounded, because formally differentiating (3) repeatedly yields

$$F^{(m)}(x) = 2^{\binom{m+1}{2}}\, F(2^m x).$$

So each summation (discrete integration) introduces the reciprocal of that growth, hence the normalization in (1).

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2) Why the runs of zeros? how long? what do they mean?

Because $\tau(2n) + \tau(2n+1) = 0$, we get immediately for the first integral

$$S^{(1)}(2m+1) = 0 \qquad \text{(zeros at all odd indices).}$$

Inductively, using $\Delta S^{(k)} = S^{(k-1)}$, we obtain the following exact statement.

> **Dyadic zero runs.**
> For every $k \geq 1$ and every $r \geq k$,
>
> $$S^{(k)}(2^r - k) = S^{(k)}(2^r - k + 1) = \cdots = S^{(k)}(2^r - 1) = 0.$$
>
> In words: *just before every dyadic endpoint $2^r - 1$ there are exactly k consecutive zeros in the k-fold sum.*

Proof sketch: for $k = 1$ we have $\sum_{j=0}^{2^r-1} \tau(j) = 0$ (in any block of length 2^r there are equally many $+1$ and -1). If $S^{(k-1)}$ vanishes on the last $k-1$ sites of such a block, then $\Delta S^{(k)} = S^{(k-1)}$ shows that $S^{(k)}$ is constant there; since $S^{(k)}(2^r - 1) = \sum_{j=0}^{2^r-1} S^{(k-1)}(j) = 0$, that constant is 0.
These zero runs are the *discrete* analogue of the fact that all derivatives of F vanish at 0 (“flatness”): at level k, the first k forward differences at dyadic endpoints are zero.

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3) “How close is it to F after N summations?”

Let $h = 2^{-k}$. On each interval $[x_j, x_{j+1}]$ the slope of F_k is the forward difference quotient. Using (2) and a Taylor expansion of F we get the uniform *local truncation error*

$$\left| \frac{F_k(x_{j+1}) - F_k(x_j)}{h} - F^{'}(x_j) \right| \ \leq \ C\, h, \qquad C = \tfrac{1}{2} \|F^{''}\|_\infty \leq 4,$$

because $F^{''}(x) = 8F(4x)$ and $0 \leq F \leq 1$ on $[0, 1]$.
A standard stability argument for this Volterra-type equation (or just summing the errors) then gives a **global uniform error**

$$\boxed{\|F_k - F\|_\infty \ \leq \ C^{'}\, 2^{-k} \quad \text{for some absolute } C^{'} \text{ (one can take } C^{'} \approx 5).} \tag{4}$$

So, after N summations (i.e. $k = N$) and the rescaling (1), the polygonal curve differs from F by $O(2^{-N})$ uniformly on $[0, 1]$.
Empirically this is what your plots show: the profiles rapidly lock onto the Fabius shape, and the visible differences are grid-scale artifacts.

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4) A compact formula for the k-fold sums

It is sometimes helpful to write the k-fold sum as a convolution with the discrete B-spline kernel

$$B_k(n) = \binom{n+k-1}{k-1} \mathbf{1}_{\{n \geq 0\}}, \qquad S^{(k)}(n) = \sum_{m=0}^{n} B_k(n-m)\, \tau(m). \tag{5}$$

The generating function viewpoint is also enlightening:

$$\sum_{n\geq 0} \tau(n) z^n = \prod_{j\geq 0}(1 - z^{2^j}), \qquad \sum_{n\geq 0} S^{(k)}(n) z^n = \frac{1}{(1-z)^k} \prod_{j\geq 0}(1 - z^{2^j}). \tag{6}$$

The factor $\prod_j (1 - z^{2^j})$ forces the cancellations responsible for the zero runs; the pole $(1-z)^{-k}$ is the k-fold integration.

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5) Asymptotics of $F(x)$ as $x \downarrow 0$ and the Lambert W function

A sharp small-x estimate drops out of the same discrete picture.

From (5), near the left edge of a dyadic block the dominant contribution to $S^{(k)}$ comes from the *leftmost* binomial weights. After the rescaling (1), a robust (and provable) proxy is

$$F(x) \approx \max_{n\geq 1} \frac{(2^n x)^n}{n!\, 2^{\binom{n}{2}}}, \qquad x \downarrow 0. \tag{7}$$

Maximizing the logarithm via Stirling ($\log n! = n \log n - n + O(\log n)$) gives

$$\Phi(n) := \log \frac{(2^n x)^n}{n!\, 2^{\binom{n}{2}}} \approx \frac{\ln 2}{2} n^2 + n \ln x - n \ln n + n.$$

Setting $\Phi'(n) = 0$ yields

$$n \ln 2 + \ln x = \ln n \quad \Leftrightarrow \quad n\, 2^{-n} = x. \tag{8}$$

Equation (8) is solved in closed form by the **Lambert W function**:

$$\boxed{n_*(x) = -\frac{1}{\ln 2} W_{-1}(-(\ln 2) x), \quad x \downarrow 0,} \tag{9}$$

where the -1 branch is the one that grows as $x \to 0^+$. Plugging $n = n_*(x)$ back into $\Phi(n)$ gives the principal asymptotics

$$\boxed{\log F(x) = -\frac{1}{2 \ln 2} W_{-1}(-(\ln 2) x)^2 + O(\log(-W_{-1}(-(\ln 2) x))), \qquad x \downarrow 0.} \tag{10}$$

Using the standard expansion $W_{-1}(-\varepsilon) = \log \varepsilon - \log |\log \varepsilon| + o(1)$ as $\varepsilon \downarrow 0$, (10) becomes the simpler and widely quoted form

$$\boxed{\log F(x) = -\frac{(\log(1/x))^2}{2 \log 2} + O(\log(1/x)\, \log\log(1/x)), \qquad x \downarrow 0.} \tag{11}$$

Thus $F(x)$ decays faster than any power of x but slower than $\exp(-c/x)$; more precisely it has the "log–squared" decay characteristic of dyadic self-similarity.